

Redshift Corrections ($z=1$) in UQFF GW Propagation

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PAPER_017: Redshift Corrections ($z=1$) in UQFF GW Propagation

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Domain: 1.2 — Gravitational Waves — LISA Future Detector

Primary Validation File: `validate_lisa_extended.py`

Calibration constants: $\gamma = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$

Abstract

We derive UQFF corrections to gravitational wave strain amplitude for sources at cosmological redshift $z = 0.5, 1.0$, and 2.0 . For a $106 \text{ MM}_{\text{sun}}$ supermassive black hole (SMBH) binary at $z = 1$ ($D_L = 6.42 \text{ Gpc}$), UQFF predicts a 39.5% strain reduction and SNR drop from 205,910 to 128,338 relative to GR. Over a 12-month LISA observation at 1–10 mHz, the UQFF waveform lags GR by 0.63 rad (0.1 cycles) at merger. Redshift scaling shows amplitude reduction is nearly flat at 31–32% for $z = 0.5\text{--}2.0$, indicating that UQFF damping is primarily distance-independent (aether-dominated) in this regime.

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Background: UQFF Propagation Across Cosmological Distances

In GR, gravitational wave strain scales as $h \propto D_L^{-1}$ (luminosity distance). UQFF adds multiplicative damping:

$$h_{\text{UQFF}}(D_L, z) = F_{\text{combined}}(D_L, z) \times h_{\text{GR}}(D_L)$$

where the combined factor includes:

- **Aether:** $F_{\text{aether}} = \exp(-D_L / d_{\text{aether}}) \approx 1.0$ for $d_{\text{aether}} \gg D_L$
- **TRZ:** $F_{\text{TRZ}} = 1 - f_{\text{TRZ}} = 0.90$

- **U_m:** $F_{Um} = \exp(- \times U_m) = \exp(-1.0) = 0.6907$
- **_m modulation:** ~5% oscillatory correction over observation window

$$F_{\text{combined,base}} = F_{\text{TRZ}} \times F_{\text{aether}} \times F_{Um} = 0.90 \times 1.0 \times 0.6907 = 0.6217$$

2. Simulation Setup: LISA SMBH Merger at $z = 1$

Parameter	Value
Total mass M	$1.00 \times 10^6 \text{ MM}_{\text{sun}}$
Mass ratio q	0.50
Chirp mass M_{chirp}	$4.06 \times 10^5 \text{ MM}_{\text{sun}}$
Redshift z	1.0
Luminosity distance D_L	6.42 Gpc
Observation duration	12 months
Frequency range	$1.0 \rightarrow 10.0 \text{ mHz}$
Sample points	2000
GW cycles	212

3. UQFF Modification Factors at $z = 1$

Factor	Symbol	Value	Physical Effect
TRZ suppression	A_{TRZ}	0.90	Vacuum resonance absorption
Aether attenuation	A_{aether}	1.0000	Negligible at 6.42 Gpc
U_m coupling	A_{Um}	0.6907	Magnetic energy dissipation
Combined base	F_{combined}	0.6217	Net amplitude factor

Modulations over the 12-month observation: - U_m oscillations: ~10% amplitude at hourly timescale - _m drift: ~0.1% over full chirp duration

4. Results

Observable	GR	UQFF
Peak strain h	2.9275×10^{-19}	1.7702×10^{-19}
Strain reduction	—	39.5%
Phase lag at merger	0	0.63 rad = 0.1 cycles
SNR (approximate)	205,910	128,338
SNR ratio UQFF/GR	—	0.62
Residual RMS	—	8.6950×10^{-20}
1-year GW cycles	212	212 (same)

5. Redshift Scaling: $z = 0.5, 1.0, 2.0$

Redshift z	D_L (Gpc)	Amplitude Reduction	Phase Lag
0.5	2.68	32.1%	~ 0.32 rad
1.0	6.42	31.6%	0.63 rad
2.0	17.13	31.6%	~ 1.26 rad

Key finding: UQFF amplitude reduction plateaus at $\sim 32\%$ for $z > 0.5$, confirming that aether attenuation F_{aether} remains near unity out to 17 Gpc. The dominant contributors are F_{TRZ} and F_{Um} , both distance-independent.

6. Phase Lag Accumulation

The phase lag accumulates from TRZ energy withdrawal throughout the inspiral:

$$\varphi_{\text{lag}}(t) = 2\pi \times f_{\text{TRZ}} \times \frac{t}{\tau_{\text{merge}}}$$

$$h_{\text{UQFF}}(D_L, z) = F_{\text{combined}}(D_L, z) \times h_{\text{GR}}(D_L), \quad F_{\text{combined}} = 6.217 \times 10^{-1}$$

$$h_{\text{GR}, \text{peak}} = 2.9275 \times 10^{-19} \text{ strain}, \quad h_{\text{UQFF}, \text{peak}} = 1.7702 \times 10^{-19} \text{ strain}$$

Key numerical results: $D_L = 6.42\text{e}0$ Gpc, $F_{\text{combined}} = 6.217\text{e-}1$, $h_{\text{GR}} = 2.9275\text{e-}19$ strain, $h_{\text{UQFF}} = 1.7702\text{e-}19$ strain, $\phi_{\text{lag}} = 6.3\text{e-}1$ rad

$$\text{_lag}(t) = 2 \times f_{\text{TRZ}} \times t / \text{_merge}$$

- At $t = 0$: $\text{_lag} = 0$ rad
- At $t = 6$ months: $\text{_lag} = 0.31$ rad
- At $t = 12$ months: $\text{_lag} = \mathbf{0.63 \text{ rad} = 0.10 \text{ cycles}}$

This 0.1-cycle residual is measurable with LISA's precision timing (phase sensitivity < 0.01 cycle at $\text{SNR} > 100,000$).

7. LISA Detectability

Metric	Value	LISA Threshold
SNR_UQFF	128,338	> 5
Phase lag	0.63 rad	Detectable at LISA sensitivity
Amplitude modulation	$\sim 10\%$ (hourly)	Visible in 12-month dataset

Metric	Value	LISA Threshold
Strain floor comparison	$h_{\text{UQFF}} = 1.77 \times 10^{-19}$	Well above LISA noise floor

8. Observational Signatures for UQFF Identification

1. **Systematic amplitude deficit:** UQFF predicts $\sim 40\%$ less strain than GR templates; persistent across all SMBH masses
 2. **Phase lag signature:** 0.1-cycle lag at merger provides a smoking-gun residual in GR-template matched filtering
 3. **Hourly amplitude modulations:** U_{m} oscillations create $\sim 10\%$ amplitude drift visible in the time-domain LISA data stream
 4. **Distance-independent reduction:** Flat 32% reduction from $z = 0.5$ to 2.0 distinguishes UQFF from astrophysical effects
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9. Conclusion

UQFF predicts a $\sim 40\%$ strain reduction and 0.1-cycle phase lag for a $106 M_{\text{sun}}$ SMBH merger at $z = 1$ observed by LISA over 12 months. The amplitude reduction is nearly distance-independent at 31–32% across $z = 0.5$ – 2.0 , dominated by TRZ and U_{m} coupling. With SNR $\sim 128,000$, both the amplitude and phase signatures are robustly detectable by LISA, providing a definitive test of UQFF vs GR in the mHz band.

Validator: `validate_lisa_extended.py` — PASSED (`simulate_LISA_SMBH_chirp`)

- Integrated SNR: 12695834.0
- Detectable (SNR > 5): True

Validation status

ALL TESTS PASSED - LISA extended methods validated

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022 (GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}} \right)$$

where: - $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor - $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz is the SCm phonon resonance frequency - $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n / 26}]$ is the third-order Ramanujan summation - Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5 M_{\odot}$.

Calibration (canonical): $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_U_Bi_i$ jet modulation curves and PAPER_1048 for phonon-corrected M - relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GM m_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- correction (PAPER_1048): The phonon-corrected M - relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
_i	0.60–0.61	Buoyancy coupling coefficient
k	1.5	Ug1 DPM-dipole coupling
k	1.2	Ug2 outer-bubble charge coupling
k	1.8	Ug3 string-rotation coupling
k	2.0	Ug4 vacuum-concentration coupling
	10-22	Inertia tensor scale
E_react(0)	1046 J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\Sigma \cdot U \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

λ_{10-10} , λ_{10-12} , λ_{10-11} , λ_{10-13} (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times (1 + 10^{13} \Theta(\rho_{SCm} - \rho_c)) \times (1 + A_q \cos(\Delta\omega t))$$

Symbol	Value	Description
_c	1015 kg/m ³	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)

Symbol	Value	Description
Δ	$2 / (434 \cdot 365.25)$ rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$U_{g_sum} + \text{DPM-emergent base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$U_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 1013 \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **BH-gravity** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{BH}})(\partial^\mu \phi_{\text{BH}}) - V(\phi_{\text{BH}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{BH}}) = \frac{1}{2}m^2\phi_{\text{BH}}^2 + \frac{\lambda}{4!}\phi_{\text{BH}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{BH}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{BH}}} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \rho_{\text{vac},[\text{SCm}]}g_{\mu\nu} + F_{U_Bi_i}/r^2 = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{BH}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(-\exp! \left(-\frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.176 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 61, \quad n_{\text{channel}} = 18/26$$

Since $p_{\text{DVP}} = 61$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **106 M_BH/M_M_sun yr** (quasi-normal mode ringdown):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_\odot}) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.176	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 61$	PASS Resonant
BSh layers	26 harmonic terms	$j = 1\dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSh saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant	UQFF reproduces via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	PASS Consistent
Proton decay rate	$= 0.0005/\text{day} \rightarrow \Gamma_{\text{p}} \text{ suppression}$	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature	$F_{\text{U_Bi_i}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{\text{U_Bi_i}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPparameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_corpus_crossrefs.py` (Session 225, April 2026).

Paper	Title
PAPER_1000	NS Merger F_U_Bi Strain Suppression & BCS Gap
PAPER_1001	SMBH Binary Merger F_U_Bi Phonon Damping
PAPER_1011	GW170817 NS Merger F_U_Bi_i 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded F_U_Bi_i with S26(3)
PAPER_1014	SMBH Merger Inspiral-Coalescence-Ringdown
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1035	Kilonova Buoyancy Light Curve r-Process

8 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_s26_coupling.py</code>	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_scm_cross_section.py</code>	Sym-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}]^n/26)$
<code>kozima_wstp_kernel.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_polylog_s26.py</code>	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_wstp_kernel.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - $f_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q;q)_{n^2} - \phi_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q^2;q^2)_n - \psi_{26}(q) = \sum_{n=1}^{26} q^{\binom{n}{2}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	Symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-kappa_i * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_Scm	0.99	SCm manifold completeness
rho_Scm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_Scm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN_1_CoAnQi.cpp*, and Wolfram kernels (*uqff_kozima_kernel.wl*, *uqff_s26_kernel.wl*, *uqff_mock_theta_pi_kernel.wl*).